# FinGraph AI: Research & Implementation Audit Report

**Target Title:** *"FinGraph AI: A Temporal Knowledge Graph and Hybrid GraphRAG Framework for Multi-Hop Financial Question Answering"*

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| **Target Venue:** IEEE / ACM Conference (Research Track) | **Audit Date:** October 2026 |
| **Auditor:** Lead Systems Architect & Senior Peer Reviewer | **Git Commit:** bcd08a2 (main / methodology-v2) |
| **Research Maturity:** 4.5 / 10 | **Implementation Maturity:** 4.0 / 10 \| **Publication Readiness:** 3.0 / 10 |

***EXECUTIVE AUDIT SUMMARY:*** *The repository underwent a major strategic pivot from an earlier stock return prediction model (archived in archive/old_ml_experiments) to a* ***Temporal Knowledge Graph & Hybrid GraphRAG QA framework****. The upstream document ingestion, structure-aware chunking, vector indexing (Qdrant), and batch LLM extraction pipelines are* ***production-ready and verified*** *with 40 passing tests. Over 96 SEC filings are processed into 38,010 entities and 9,448 relations.* ***CRITICAL FINDING:*** *The downstream Graph Retrieval, Hybrid Fusion, Cross-Encoder Reranking, QA Agent, Evaluation Suite, and Benchmark Question Dataset are* ***currently empty (0 bytes)****. Furthermore, an Entity ID mismatch between extraction (hashing) and graph building (name-string) prevents extracted triples from forming edges. The paper cannot be submitted until these core components are implemented.*

## 1. Current Implementation Status

A line-by-line inspection of all repository modules was conducted to verify actual code existence versus architectural documentation:

| Subsystem / Component | Module / File Path | Status | Lines | Readiness / Notes |
|---|---|---|---|---|
| **SEC Ingestion** | src/data/ingestion/sec_corpus.py sec_content.py | **COMPLETE** | 663 | Downloads 10-K, 10-Q, 8-K; cleans HTML, tracks EDGAR acceptance timestamps. |
| **Chunking Engine** | src/data/chunking/structure_aware.py | **COMPLETE** | 178 | Paragraph-aware boundaries, bounded sliding overlap, preserves temporal metadata. |
| **Entity/Relation Extraction** | src/data/extraction/llm.py pipeline.py, entities.py | **COMPLETE** | 1,052 | Ollama & OpenRouter support. 11 entity & 13 relation types. 96 filings extracted. |
| **Temporal Graph Builder** | src/data/graph/builder.py schema.py, snapshot.py | **PARTIAL / BUG** | 642 | **Critical ID mismatch bug:** Drops extracted relations. Only legacy AAPL graph built. |
| **Vector Retrieval** | src/retrieval/vector.py scripts/build_vector_index.py | **COMPLETE** | 243 | Qdrant store with available_time <= reference_time filter. Fully tested. |
| **Graph Retrieval** | src/retrieval/graph.py | **MISSING** | 0 | Empty file. No entity linking, BFS traversal, or subgraph retrieval. |

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| **Hybrid Retrieval & Fusion** | src/retrieval/hybrid.py | **MISSING** | 0 | Empty file. No Reciprocal Rank Fusion (RRF) between vector & graph. |
| **Cross-Encoder Reranker** | src/retrieval/reranker.py | **MISSING** | 0 | Empty file. No cross-encoder reranking implemented. |
| **Multi-Hop QA & Agent** | src/qa/agent.py, generation.py query_analysis.py | **MISSING** | 0 | Empty files. No query decomposition, synthesis, or citation generation. |
| **Benchmark Dataset** | benchmark/questions.json | **MISSING** | 0 | Empty file. Zero evaluation questions or ground-truth annotations. |
| **Evaluation Pipeline** | src/evaluation/* scripts/run_evaluation.py | **MISSING** | 0 | Empty files. Zero metrics implemented (HitRate, F1, Faithfulness, Leakage). |
| **Unit & Regression Tests** | tests/test_*.py (6 test files) | **COMPLETE** | 1,046 | **40 passed in 77s.** Covers extraction, chunking, snapshots, vector store. |

## 2. Research Methodology Audit

We trace the intended scientific lifecycle: *Document → Chunking → Temporal Metadata → Extraction → Temporal KG → Hybrid Retrieval → Reranking → Multi-Hop QA → Temporal Validation → Provenance*

- **Document Ingestion:** **[COMPLETE]** — SEC filings for 10 Fortune 500 corporations (AAPL, MSFT, NVDA, GOOGL, AMZN, META, TSLA, JPM, JNJ, XOM) are cleanly ingested and normalized.
- **Structure-Aware Chunking:** **[COMPLETE]** — Splits along semantic boundaries; tables and paragraphs are kept intact with bounded sliding character fallback.
- **Temporal Metadata Tagging:** **[COMPLETE]** — Separates event_time, publication_date, available_time (acceptance datetime), and fiscal_period.
- **Entity & Relation Extraction:** **[COMPLETE]** — High quality extraction over 96 filings yielding 38,010 entities and 9,448 relations with confidence scoring.
- **Knowledge Graph Assembly:** **[PARTIALLY COMPLETE]** — Schema exists, but extracted triples are unlinked due to an entity-ID mismatch. Multi-company KG is not yet saved to disk.
- **Vector Retrieval:** **[COMPLETE]** — Indexed in Qdrant with all-MiniLM-L6-v2; point-in-time temporal filtering is verified and working.
- **Graph Retrieval:** **[MISSING]** — No code exists to traverse nodes, find multi-hop connections, or return contextual triples.
- **Hybrid Retrieval (Fusion):** **[MISSING]** — No rank fusion (e.g., RRF) exists to unite vector text chunks with relational graph triples.
- **Reranking:** **[MISSING]** — No cross-encoder model to score and re-order candidate evidence.
- **Multi-Hop Reasoning:** **[MISSING]** — No multi-step planner or chain-of-thought engine to traverse multi-document financial paths.
- **Temporal Validation:** **[NEEDS IMPROVEMENT]** — Point-in-time logic exists in isolation (vector query filter, snapshot script), but is not integrated into end-to-end QA.
- **Evidence & Provenance:** **[NEEDS IMPROVEMENT]** — Metadata payload contains chunk URLs and accession numbers, but no citation generator exists to cite them in answers.

# 3. Identification of Actual Novelty

To withstand strict peer review at top AI/Data conferences, claims of generic superiority must be replaced with defensible, mathematically grounded contributions:

## A. What is NOT Novel in FinGraph AI (Do NOT claim as novelty):

- Chunking financial documents and storing vectors in Qdrant (Standard Vector RAG).
- Using an LLM to extract (entity, relation, entity) triples (Standard Open-IE / GraphRAG).
- Merging vector search with graph neighbors (Prior art: Microsoft GraphRAG, HippoRAG, HybridRAG).

## B. What IS Genuinely Defensible as Novel:

1. **Bi-Temporal Epistemic Decoupling in Financial KG:** Existing Temporal KGQA frameworks (e.g., CronKGQA) model edges as (s, r, o, [t_start, t_end]). In corporate reporting, there is a fundamental gap between *event_time / fiscal_period* (when the quarter closed) and *available_time* (when EDGAR made the filing public). FinGraph AI enforces **point-in-time epistemic availability** across both vector embeddings and relational graph edges.
2. **Controlled-Ontology Multi-Hop Cross-Filing Grounding:** Unlike generic open-IE graphs that suffer from relation explosion and noisy hallucinations, FinGraph AI restricts extraction to 11 entity types and 13 structural business relationships (SUPPLIES, DEPENDS_ON, COMPETES_WITH, HAS_RISK). This allows deterministic multi-hop traversal connecting supply chain disruptions across distinct corporate filers.

# 4. Research Weaknesses & IEEE Reviewer Critique

Simulating a strict IEEE reviewer evaluation, the following critical vulnerabilities were identified that would cause immediate paper rejection if not resolved:

- **Fatal Flaw: Architectural Ghost Code:** The paper title promises a 'Hybrid GraphRAG Framework for Multi-Hop Financial QA', but retrieval, reranking, hybrid fusion, and QA generation are completely unwritten (0 bytes). Reviewers will desk-reject an empirical paper without an operational pipeline.
- **Fatal Flaw: Missing Evaluation Benchmark:** benchmark/questions.json is empty. There is zero empirical evidence comparing FinGraph AI against Vector RAG or GraphRAG baselines. A paper cannot be submitted without quantitative tables (Recall, NDCG, F1, Faithfulness).
- **Critical Defect: Pipeline Disconnection (Entity ID Bug):** Extraction produces IDs like company:5d677c32e599568a (SHA-256 hash), while the graph builder constructs IDs like company:apple. Furthermore, relations use source_entity_id while builder expects source. If run today, 100% of the 9,448 extracted relations are dropped as dangling.
- **Methodology Vulnerability: Casing Mismatch in Snapshot:** In src/data/graph/snapshot.py, line 105 checks node.node_type == 'company'. The semantic extraction ontology uses 'Company'. Consequently, company nodes without explicit available_time are dropped from point-in-time snapshots.
- **Missing Baseline Suite:** The system fails to benchmark against: (1) Dense Vector-only RAG, (2) BM25+Vector Hybrid, (3) Atemporal GraphRAG, and (4) Graph-only retrieval. Reviewers require comparative proof of incremental component utility.

# 5. Prioritized Action Plan

## P0 — Essential Before Paper Submission (Research Validity)

- **Fix Entity ID Resolution & Build Semantic KG** [src/data/graph/builder.py, scripts/build_semantic_kg.py, Difficulty: Low-Med] — Harmonize entity IDs between extraction (hashes) and builder (names). Compile the 96 extracted JSONs into a unified multi-company graph.
- **Implement Temporal Graph Retrieval** [src/retrieval/graph.py, Difficulty: Medium] — Extract query entities, traverse k-hop temporal subgraphs (available_time <= t_q), linearize triples into text.
- **Implement Hybrid Rank Fusion (RRF)** [src/retrieval/hybrid.py, Difficulty: Low] — Merge vector chunk rankings with graph triple evidence using Reciprocal Rank Fusion.
- **Implement QA Generation Agent** [src/qa/generation.py, src/qa/agent.py, Difficulty: Medium] — Prompt LLM with fused context, enforce formal citations [Doc, Chunk, AvailableTime].
- **Curate Financial QA Benchmark** [benchmark/questions.json, Difficulty: Medium] — Annotate 100–120 verified single-hop, multi-hop, and temporal financial questions.

## P1 — Strong Research Improvements (Acceptance Drivers)

- **Cross-Encoder Reranker** [src/retrieval/reranker.py, Difficulty: Low-Med] — Add ms-marco-MiniLM or BGE-reranker to boost NDCG@5 by 5–12%.
- **Automated Evaluation Suite** [src/evaluation/*, scripts/run_evaluation.py, Difficulty: Medium] — Implement Recall@K, MRR@K, Token F1, Faithfulness (RAGAS style), and Temporal Leakage Violation Rate (TLVR).

## P2 — Deprioritized (Do NOT Waste Time Before Submission)

- **Do NOT build a Web UI:** Reviewers evaluate empirical rigor, not frontend dashboards.
- **Do NOT revive stock price prediction ML:** Archived in old_ml_experiments; out of scope for QA paper.
- **Do NOT add Macro/News APIs:** 99 SEC 10-K/Q filings provide more than enough density.

# 6. Evaluation Design & Ablation Framework

To provide incontrovertible scientific evidence, the experimental evaluation is structured across 5 query categories, 5 baseline systems, and 4 ablation variants:

| Query Category | Count | Target Capability Tested | Key Metric |
|---|---|---|---|
| **Single-Hop Factoid** | 25 | Direct factual retrieval from single filing sections | Recall@3, HitRate@1 |
| **Multi-Hop Relational** | 30 | Cross-company supply chain / dependency traversal across filings | Multi-Hop Recall, Token F1 |
| **Temporal Point-in-Time** | 25 | Epistemic cutoff validation (verifying knowledge at time t_q) | Temporal Violation Rate (TLVR) |
| **Comparative Analysis** | 20 | Cross-company metric & risk comparison in shared fiscal windows | Factual Accuracy, ROUGE-L |
| **Temporal Aggregation** | 20 | Multi-quarter financial trend synthesis across successive 10-Qs | Faithfulness, Citation Precision |

### Ablation Configurations to Execute:

- **A1: Full FinGraph AI:** Vector + Temporal KG + RRF + Cross-Encoder Reranker + Provenance.
- **A2: w/o Knowledge Graph (Vector Only):** Proves the statistical lift of relational paths.
- **A3: w/o Temporal Validation (Atemporal GraphRAG):** Quantifies future leakage (TLVR > 0).
- **A4: w/o Cross-Encoder Reranker:** Measures the precision contribution of the reranking stage.
- **A5: w/o Multi-Hop Expansion (k=1 vs k=2):** Demonstrates gain from 2-hop relational traversal.

## 7. Temporal Leakage & Epistemic Correctness Audit

A strict mathematical audit was performed on temporal isolation: *t_query → Knowledge(t ≤ t_query) → Retrieval → Reasoning → Answer*

**Strengths:** SEC EDGAR acceptance datetimes are accurately recorded as available_time. Qdrant vector filtering strictly enforces available_time <= reference_time. Graph snapshots enforce that both nodes and the connecting edge must satisfy availability.

**Vulnerability 1: Case Sensitivity Bug:** src/data/graph/snapshot.py line 105 tests lowercase node.node_type == 'company' while extraction outputs 'Company', causing company nodes to drop from historical snapshots unless patched.

**Vulnerability 2: Query-Time Leakage:** The missing QA agent currently lacks an automatic temporal date parser. If a question does not explicitly inject reference_time, Qdrant defaults to unconstrained search, retrieving future filings.

## 8. DO THIS NEXT — Recommended Execution Sequence

To transform this repository into an IEEE-ready research submission, proceed in this exact chronological order:

**STEP 1: Reconcile Identifiers & Compile Semantic KG:** Fix ID mismatch in extraction/builder; patch snapshot casing; run script to compile the 96 extracted files into fingraph_semantic_kg.json.

**STEP 2: Implement Graph & Hybrid Retrieval Modules:** Write src/retrieval/graph.py (k-hop expansion) and src/retrieval/hybrid.py (RRF fusion) + tests.

**STEP 3: Implement QA Generation & Baselines:** Write src/qa/generation.py (citation prompt), scripts/run_rag_baseline.py, and scripts/run_fingraph.py.

**STEP 4: Annotate Evaluation Benchmark:** Populate benchmark/questions.json with 100–120 verified financial questions with gold evidence IDs.

**STEP 5: Run Evaluation Suite & Generate Paper Tables:** Implement src/evaluation/*; run full baseline and ablation suite; output publication-ready LaTeX tables.